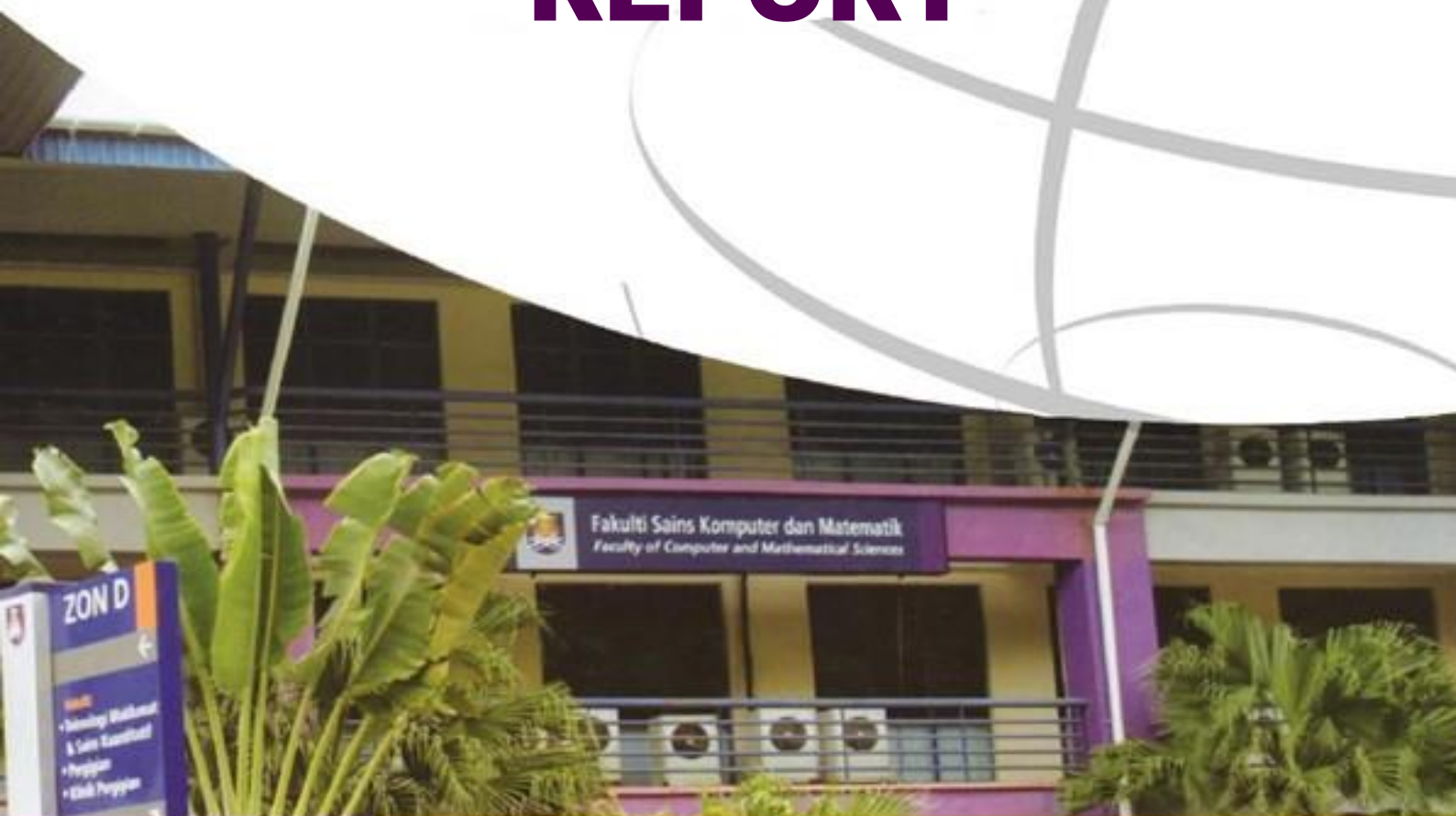




اُنِيْوَرْسِيْٓتِيْ تِكْنُوْلُوْجِيْ مَآرَا
UNIVERSITI
TEKNOLOGI
MARA

Faculty of Computer and Mathematical Sciences

PRACTICAL TRAINING REPORT



INDUSTRIAL TRAINING REPORT

at

BATERIKU (M) SDN. BHD.

SHARIFAH NUR AIN ASYIAH BINTI SYED OMAR

2023190329

UNIVERSITI TEKNOLOGI MARA

INDUSTRIAL TRAINING REPORT

At

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Jelutong, 40150
Shah Alam, Selangor**

17 September 2025 to 19 December 2025

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2023190329**

**Report submitted in partial fulfillment of the requirement
for the degree of
Bachelor of Computer Science (Hons.) Multimedia Computing**

Faculty of Computer and Mathematical Sciences

DECEMBER 2025

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Date: 30 December 2025

Name: Sharifah Nur Ain Asiyah Binti Syed Omar

TABLE OF CONTENTS

	Page
ACKNOWLEDGEMENTS	1
TABLE OF CONTENTS	2
LIST OF FIGURES	4
CHAPTER 1: INTRODUCTION	7
1.1 Background of Industrial Training	7
1.2 Background of Organization	7
1.3 Background of Department Attached	8
CHAPTER 2: ASSIGNMENTS	10
2.1 Task 1: Website Enhancement for bateriku.com	10
2.1.1 Introduction	10
2.1.2 Implementation	10
2.1.3 Result and Discussion	13
2.2 Task 2: Content Update for business.bateriku.com	16
2.2.1 Introduction	16
2.2.2 Implementation	16
2.2.3 Result and Discussion	16
2.3 Task 3: Structured Data Updates for bateriku.edu.my	17
2.2.1 Introduction	17
2.2.2 Implementation	17
2.2.3 Result and Discussion	20
2.4 Task 4: Image Fix for corporate.bateriku.com	21
2.2.1 Introduction	21
2.2.2 Implementation	21
2.2.3 Result and Discussion	21
2.5 Task 5: Frontend Development & SEO Implementation for astraman.my	22
2.2.1 Introduction	22
2.2.2 Implementation	22
2.2.3 Result and Discussion	26

CHAPTER 3 : CONCLUSION AND RECOMMENDATIONS	29
3.1 Benefits of the Assignments	29
3.2 Summary of Industrial Training	29
BIBLIOGRAPHY	31

LIST OF FIGURES

Figure 1: Company Logo.....	7
Figure 2: Code for updating News page title and description (bateriku.com). Red: code before update; Green: code after update.....	10
Figure 3: Code for updating footer label from “News & Updates” to “Articles” (bateriku.com). Red: old label; Green: new label.....	11
Figure 4: Code fix for layout overlapping issue between the Complaint section and Battery Order Form (bateriku.com). Red: before fix; Green: after fix.....	11
Figure 5: Code for removing loading page after successful purchase submission (bateriku.com). Red: before removal; Green: after removal.....	12
Figure 6: Code for implementing location filter restricting search results to Malaysia (bateriku.com). Red: before; Green: after.....	12
Figure 7: Code for adjusting banner height from full height to screen height (bateriku.com). Red: before; Green: after.....	13
Figure 8: News page before and after content update (bateriku.com).....	13
Figure 9: Footer label before and after update (bateriku.com).....	14
Figure 10: Layout overlapping issue before and after fix (bateriku.com).....	14
Figure 11: Before removing the loading page after successful purchase submission (bateriku.com).....	15
Figure 12: Location filter restricting search results to Malaysia on bateriku.com. before and after implementation (bateriku.com).....	15
Figure 13: Banner height before and after adjustment (bateriku.com).....	15
Figure 14: Code for updating News page title and description (business.bateriku.com). Red: before; Green: after.....	16

Figure 15: News page before and after update (business.bateriku.com).....	17
Figure 16(a): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my).....	18
Figure 16(b): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my).....	18
Figure 16(c): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my).....	18
Figure 16(d): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my).....	19
Figure 16(e): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my).....	19
Figure 16(f): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my).....	19
Figure 17: Code for updating Organization structured data on Akademi Bateriku (bateriku.edu.my). Red: before; Green: after.....	20
Figure 18: Structure data schema of BEV course page after update (bateriku.edu.my).....	20
Figure 19: Structure data schema of Organization after update (bateriku.edu.my)...	21
Figure 20: Code for fixing inconsistent image display on Corporate Bateriku News page (corporate.bateriku.com). Red: before; Green: after.....	21
Figure 21: Image display before and after fix (corporate.bateriku.com).....	22
Figure 22: Icon images used for successful and unsuccessful pages (astraman.my).....	23

Figure 23: Background image used for successful and unsuccessful pages (astraman.my).....	23
Figure 24: Source code implementation for the successful page (astraman.my).....	24
Figure 25: Source code implementation for the unsuccessful page (astraman.my)..	25
Figure 26: Source code for creating the post entry in the sitemap (astraman.my)....	26
Figure 27: Source code for implementing structured data for the Product Catalogue (astraman.my).....	26
Figure 28: Unsuccessful page (astraman.my).....	27
Figure 29: Successful(thank you) page (astraman.my).....	27
Figure 30: Structured data thumbnails before and after update for the Sponsorship and Contact Us pages (astraman.my).....	27
Figure 31: Post entry in the sitemap for the News page (astraman.my).....	28
Figure 32: Structured data schema for the Product Catalogue (astraman.my).....	28

CHAPTER 1

INTRODUCTION

1.1 Background of Industrial Training

Industrial training is a compulsory requirement for final-year students enrolled in the Bachelor of Computer Science (Hons.) Multimedia Computing (CDCS253). The industrial training subject, CST688, is designed to provide students with practical experience in a real working environment related to their field of study. This training is conducted over a period of 14 weeks.

The main purpose of industrial training is to help students apply the knowledge and skills learned during their academic studies to real workplace situations. Through this training, students are exposed to actual tasks and projects, which allow them to understand better how theoretical concepts are used in the industry.

In addition to technical skills, industrial training also helps students develop professional skills such as responsibility, teamwork, discipline, and the ability to adapt to a working environment. Students also gain exposure to workplace culture, work ethics, and organizational structure. Overall, industrial training plays an important role in preparing students to become more confident and ready for employment after graduation.

1.2 Background of Organization



Figure 1: Company Logo

Bateriku.com was established in March 2014 as an on-demand car battery replacement service provider. Over the years, it has evolved into an ecosystem that connects car users to roadside assistance, trusted workshops, an auto parts marketplace, and other ancillary services.

Bateriku.com has scaled its operations by leveraging big data, technology, and the sharing economy. As of today, the company operates nearly 200 retail outlets (Pitstops) nationwide and has served more than 1.5 million car owners. Its services cover all states in Malaysia, including Sabah and Sarawak. The company also plans to expand regionally, starting with its first Pitstop in Serpong, South Jakarta, Indonesia.

In addition, Bateriku.com provides centralized training to internal staff and external partners through Akademi Bateriku, a training institution endorsed by the Department of Skills Malaysia (JPK) and the National Dual Training System (SLDN) under the Ministry of Human Resources of Malaysia. This ensures that all personnel are equipped with proper skills and knowledge to deliver high-quality services to customers.

1.3 Background of Department Attached

During my industrial training, I was placed in the Engineering Department as a Frontend Intern at Bateriku.com. This department is responsible for developing, maintaining, and improving the company's websites to ensure they are user-friendly, functional, and up to date.

As a Frontend Intern, I worked mainly using Visual Studio Code (VS Code) as the development tool and Nuxt.js as the frontend framework. My scope of work involved maintaining and updating five company websites, namely `bateriku.com`, `business.bateriku.com`, `corporate.bateriku.com`, `bateriku.edu.my`, and `astraman.my`.

I worked closely with the production team, as they were responsible for assigning tasks to me. Whenever I did not understand the task requirements, I communicated with the production team to seek clarification. In addition, I received guidance and support from my mentor, Encik Aiman, as well as other frontend team members whenever I faced technical difficulties or needed further explanation. This collaboration helped ensure that tasks were completed according to requirements and within the given timeline.

Throughout the internship, I was involved in various frontend development and website maintenance tasks. These included updating page titles and descriptions, fixing layout and display issues, improving user experience by removing unnecessary loading pages, and implementing location filters to ensure accurate search results. I also worked on updating structured data, fixing image display issues, developing confirmation pages based on Figma designs, updating sitemap entries, and creating structured data for product catalogues.

Overall, being placed in the Engineering Department provided me with valuable hands-on experience in frontend development, teamwork, and real industry workflow, which enhanced both my technical skills and understanding of professional working environments.

CHAPTER 2

ASSIGNMENTS

2.1 Task 1: Website Enhancement for bateriku.com

2.1.1 Introduction

bateriku.com is the main consumer-facing website where customers can book services like battery replacement, towing, and other roadside assistance. This task involved enhancing and maintaining the main company website, bateriku.com, to improve content accuracy, layout structure, and user experience. The tasks focused on updating page titles and descriptions, fixing layout issues, improving navigation clarity, and ensuring smoother user interaction during the purchase journey.

2.1.2 Implementation

The first implementation involved updating the News page on bateriku.com. The News page on bateriku.com was updated with a new title and description for both English and Bahasa Melayu versions to improve clarity and content accuracy.

```
app/pages/nc/rencana/index.vue
7 7 </h1>
8 8 <div class="space-y-4 mt-2">
9 9   <h2 class="text-5xl font-medium -tracking-[2px] text-bateriku-900 antialiased">Rencana</h2>
10 10   <p class="text-lg font-normal text-bateriku-900 antialiased">Jangan ketinggalan cerita terbaru
11 11   dari Bateriku - berita hangat, pencapaian hebat, kerjasama menarik, dan pelbagai pengumuman
12 12   yang membentuk masa depan inovatif kami.</p>
13 13   <p class="text-lg font-normal text-bateriku-900 antialiased">Terokai pelbagai rencana dari Bateriku - panduan penjagaan
14 14   kereta, gaya hidup, inspirasi percutian, serta kisah-kisah yang dekat dengan pemandu.</p>
15 15 </div>
16 16 <div class="h-[2px] lg:max-w-[819px] w-full mx-auto justify-self-center max-w-[480px] mt-[31px] bg-bateriku-600 border-0" />

app/pages/news/index.vue
3 3 <section class="bg-background-white lg:pt-[100px] lg:pb-[77px] xl:px-[270px] lg:px-16 px-8 py-10">
4 4   <div class="mx-auto max-w-7xl">
5 5     <div class="text-center max-w-[900px] place-self-center">
6 6     <h1 class="text-lg font-medium text-bateriku-800 uppercase antialiased tracking-[0.2em]">News</h1>
7 7     <h1 class="text-lg font-medium text-bateriku-800 uppercase antialiased tracking-[0.2em]">Article</h1>
8 8     <div class="space-y-4 mt-2">
9 9     <h2 class="text-5xl font-medium -tracking-[2px] text-bateriku-900 antialiased">News</h2>
10 10    <p class="text-lg font-normal text-bateriku-900 antialiased">Stay updated with the latest news,
11 11    milestones, collaborations, and announcements shaping the innovative future of Bateriku.</p>
12 12    <h2 class="text-5xl font-medium -tracking-[2px] text-bateriku-900 antialiased">Articles</h2>
13 13    <p class="text-lg font-normal text-bateriku-900 antialiased">Explore a wide range of articles from Bateriku - car care
14 14    guides, lifestyle tips, travel inspiration, and stories made for drivers.</p>
15 15    </div>
16 16    <div class="h-[2px] lg:max-w-[819px] w-full mx-auto justify-self-center max-w-[480px] mt-[31px] bg-bateriku-600 border-0" />
```

Figure 2: Code for updating News page title and description (bateriku.com). Red: code before update; Green: code after update.

The footer component file was modified to change the footer label from "News & Updates" to "Articles" across all pages.

```

@@ -98,7 +98,7 @@ const footerData = ref([
  98 98 link: "#coverage",
  99 99 },
  100 100 {
  101 - title: "News & Updates",
  101 + title: "Articles",
  102 102 slug: "news-updates",
  103 103 link: "#news",
  104 104 },

```

Figure 3: Code for updating footer label from “News & Updates” to “Articles” (bateriku.com). Red: old label; Green: new label.

A layout issue was resolved where the Complaint section overlapped the Battery Order Form. The issue occurred on large screens and was fixed by adjusting the section height from screen to full.

```

@@ -1,7 +1,7 @@
...
1 1 <template>
2 2 <div class="flex flex-col bg-background-white">
3 3 <div
4 - class="bg-[image:var(--bg-desktop)] bg-no-repeat bg-fixed bg-cover mt-[-104px] contain-content mx-auto w-full lg:h-screen lg:max-h-[1440px] flex lg:py-[2.5rem]" :style="bgVars">
4 + class="bg-[image:var(--bg-desktop)] bg-no-repeat bg-fixed bg-cover mt-[-104px] contain-content mx-auto w-full lg:h-full lg:max-h-[1440px] flex lg:py-[2.5rem]" :style="bgVars">
5 5 <div
6 6 class="container mx-auto lg:p-[1rem] flex lg:gap-[90px] h-min self-center lg:justify-center flex-col-reverse lg:flex-row items-center">

```

Figure 4: Code fix for layout overlapping issue between the Complaint section and Battery Order Form (bateriku.com). Red: before fix; Green: after fix.

The Bateriku Purchase Journey was streamlined by removing a loading page that appeared after form submission.

```

... 12 app/components/home/main/form.vue
... 1 <template>
... 2 <div>
... 3 <div v-if="loading"

Check warning on line 3 in app/components/home/main/form.vue
GitHub Actions / lint
Expected a linebreak before this attribute

4 4 class="absolute top-0 left-0 w-full h-full bg-white opacity-80 flex justify-center items-center z-50">
5 - <loadingForm v-if="step == 4" />
5 +
6 6 <loadingPay v-if="step >= 5" />
7 7 </div>
8 8 <!-- plate number -->

667 667
668 668 async function displayProduct() {
669 - loading.value = true;
669 +
670 670 if (reg_number.value && name.value && phone_number.value && email.value) {
671 671 const bodyProduct = {
672 672 user: {

686 686 if (temp_product?.products?.length) {
687 687 product.value = sortByRanking(temp_product.products);
688 688 product_id.value = product.value[productCount.value].id;
689 - setTimeout(() => {
689 + step.value = 5;
690 - loading.value = false;
691 - step.value = 5;
692 - }, 5000);
693 693 } else {
694 694 submitOrderLead();
695 - setTimeout(() => {
696 - loading.value = false;
697 - }, 5000);
698 698 }

```

Figure 5: Code for removing loading page after successful purchase submission (bateriku.com). Red: before removal; Green: after removal.

A location filter was implemented to restrict search results to Malaysia only. Previously, search results included locations from all countries, creating confusion for users.

```

594 594
595 595 // Initialize Autocomplete
596 - const autocomplete = new $googleMaps.places.Autocomplete(autocompleteInput.value);
596 + const autocomplete = new $googleMaps.places.Autocomplete(autocompleteInput.value, {
597 + componentRestrictions: { country: 'MY' }, // Restricts to Malaysia
598 + fields: ['formatted_address', 'geometry', 'address_components']
599 + });
600 +

597 601 autocomplete.addListener('place_changed', () => {
598 602 const place = autocomplete.getPlace();
599 603 if (!place.geometry) return;

```

Figure 6: Code for implementing location filter restricting search results to Malaysia (bateriku.com). Red: before; Green: after.

The banner section on bateriku.com required adjustment due to visual display issues. The banner was originally set to full page height in large screen size, which caused the banner image to be cut off when users zoomed out on their browsers. This created an unprofessional appearance and prevented users

from seeing the complete banner design. The CSS height property was modified from full page height to screen height.

```

1 1 <template>
2 2 <div class="flex flex-col bg-background-white">
3 3 <div
4 4 - class="bg-[image:var(--bg-desktop)] bg-no-repeat bg-fixed bg-cover mt-[-104px] contain-content mx-auto w-full lg:h-full lg:max-h-[1440px] flex lg:py-[2.5rem] " :style="bgvars">
5 5 <div
6 6 + class="bg-[image:var(--bg-desktop)] bg-no-repeat bg-fixed bg-cover mt-[-104px] contain-content mx-auto w-full lg:h-screen lg:max-h-[1440px] flex lg:py-[2.5rem] " :style="bgvars">
7 7 <div
8 8 class="container mx-auto lg:py-[1rem] flex lg:gap-[90px] h-min self-center lg:justify-center flex-col-reverse lg:flex-row items-center">
9 9 <div class="hidden lg:flex flex-col flex-start min-h-[432px] h-min content-end justify-end mb-0 mt-auto">

```

Figure 7: Code for adjusting banner height from full height to screen height (bateriku.com). Red: before; Green: after.

2.1.3 Results and Discussion

After implementation, the website displayed clearer content structure and improved visual layout. The updated titles and footer labels enhanced readability and navigation clarity. The layout fixes resolved overlapping issues, while the updated banner height improved overall page appearance across devices. The removal of the loading page also provided a smoother user experience after purchase submission.

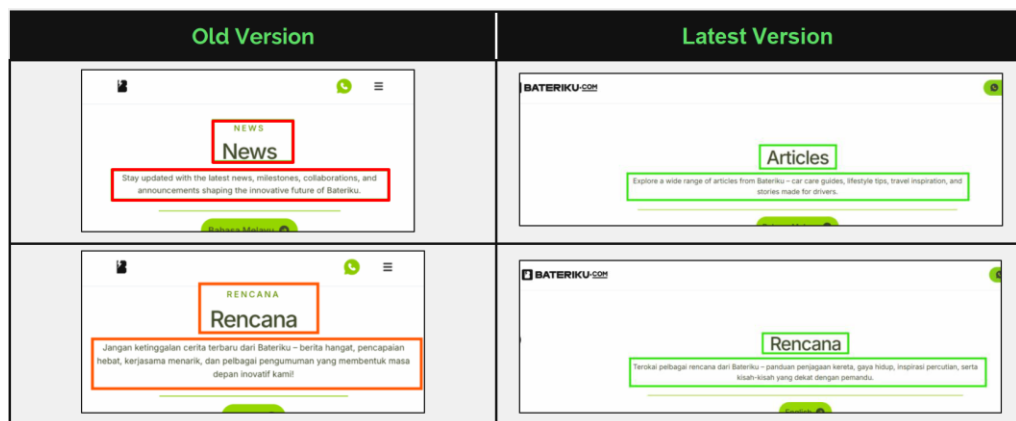


Figure 8: News page before and after content update (bateriku.com)

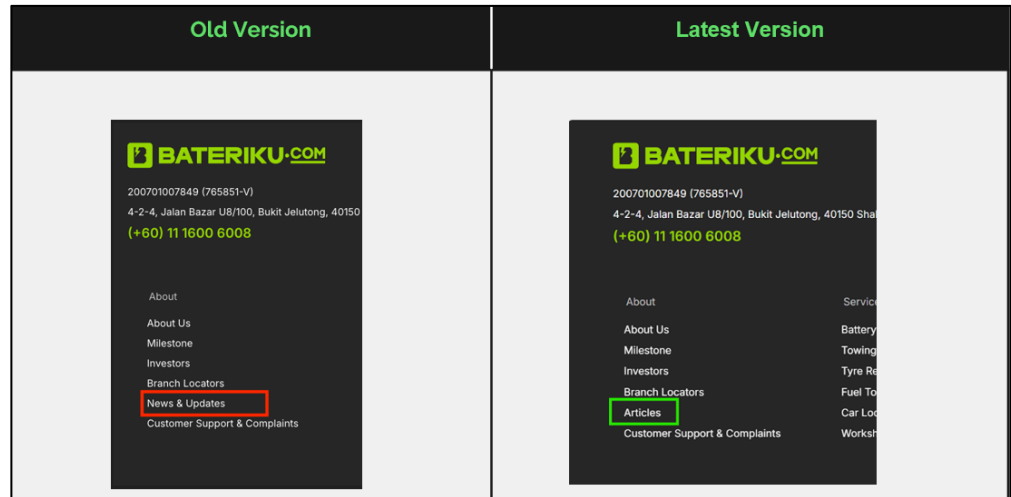


Figure 9: Footer label before and after update (bateriku.com)

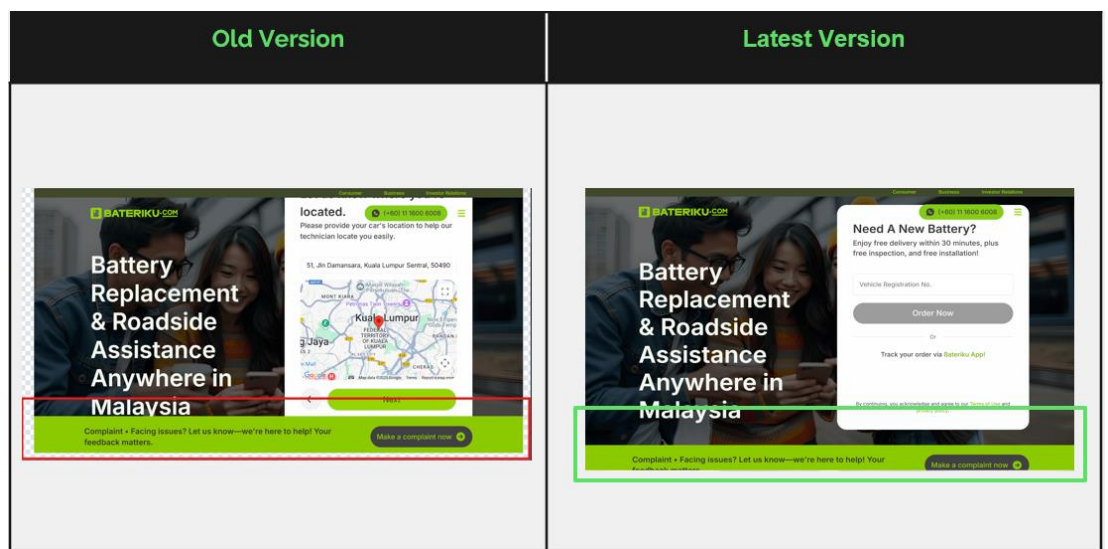


Figure 10: Layout overlapping issue before and after fix (bateriku.com)

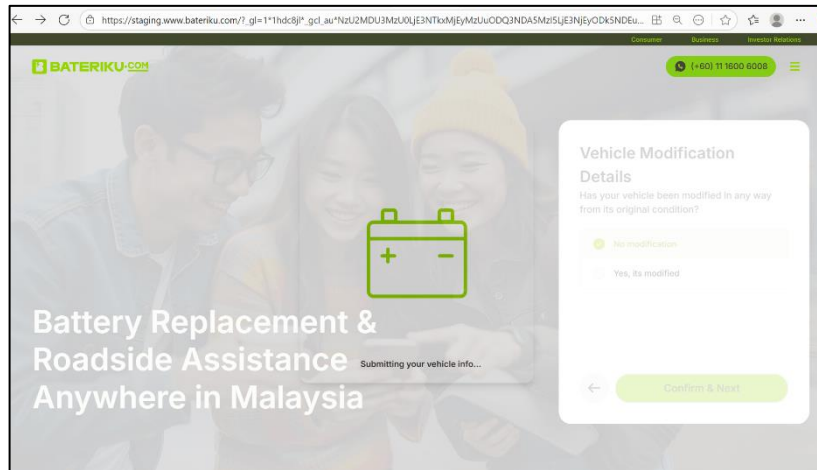


Figure 11: Before removing the loading page after successful purchase submission (bateriku.com)

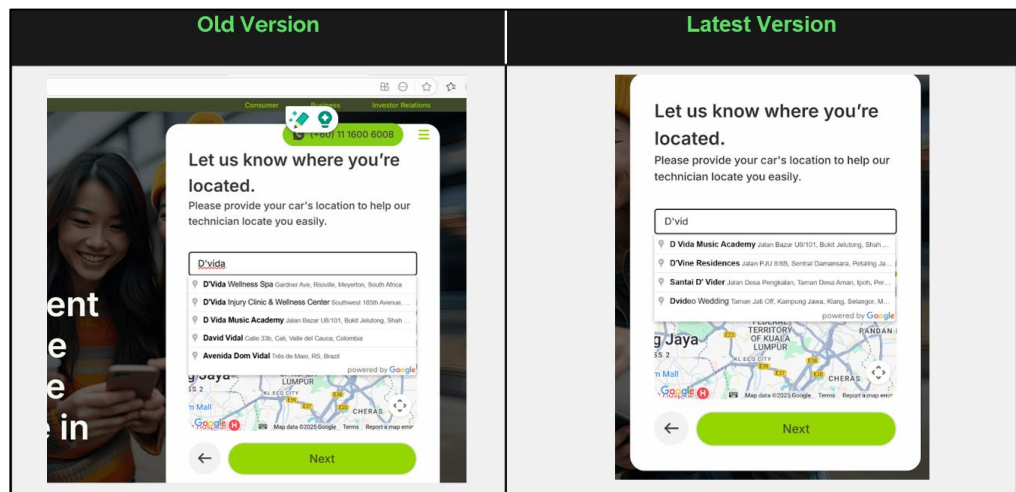


Figure 12: Location filter restricting search results to Malaysia on bateriku.com. before and after implementation (bateriku.com)

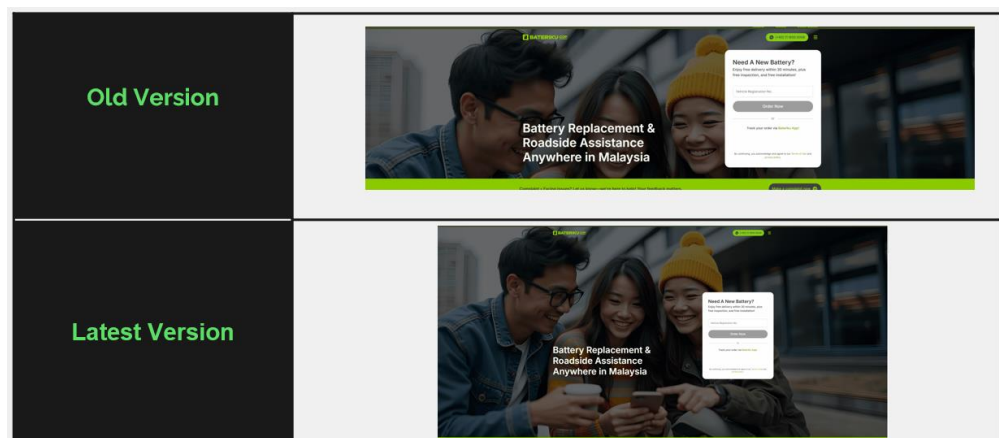


Figure 13: Banner height before and after adjustment (bateriku.com)

2.2 Task 2: Content Update for business.bateriku.com

2.2.1 Introduction

business.bateriku.com is Bateriku's business portal, providing information on digital partner and career opportunities, including on-demand automotive services, and entrepreneur programs. This task focused on updating content for business.bateriku.com, specifically to improve language accuracy and consistency for the News section.

2.2.2 Implementation

The News page content was updated by revising the news description in both English and Malay versions and changing the Malay title from “Rencana” to “Berita” to ensure proper language usage and clarity.



```
app/assets/lang/en.json
@@ -1,4 +1,4 @@
1 1 {
2 2   "News": "News",
3 3 -   "Stay updated with the latest news, milestones, collaborations, and announcements shaping the innovative future of Bateriku.":"Stay updated
with the latest news, milestones, collaborations, and announcements shaping the innovative future of Bateriku."
3 3 +   "Stay updated with the latest news, milestones, collaborations, and announcements shaping the innovative future of Bateriku.":"Stay up to
date with the latest Bateriku news - ecosystem growth, strategic collaborations, company achievements, and automotive industry highlights."
4 4 }

app/assets/lang/ms.json
@@ -1,4 +1,4 @@
1 1 {
2 2   "News": "Rencana",
3 3 -   "Stay updated with the latest news, milestones, collaborations, and announcements shaping the innovative future of Bateriku.":"Jangan
ketinggalan dengan berita terkini, pencapaian, kerjasama dan pelbagai pengumuman yang membentuk masa depan inovatif Bateriku."
2 2 +   "News": "Berita",
3 3 +   "Stay updated with the latest news, milestones, collaborations, and announcements shaping the innovative future of Bateriku.":"Ikuti berita
terkini dari Bateriku - perkembangan ekosistem, kolaborasi strategik, pencapaian syarikat, dan sorotan industri automotif."
4 4 }
```

Figure 14: Code for updating News page title and description
(business.bateriku.com). Red: before; Green: after.

2.2.3 Results and Discussion

The updated News section now shows clearer, making the content easier to understand for users.

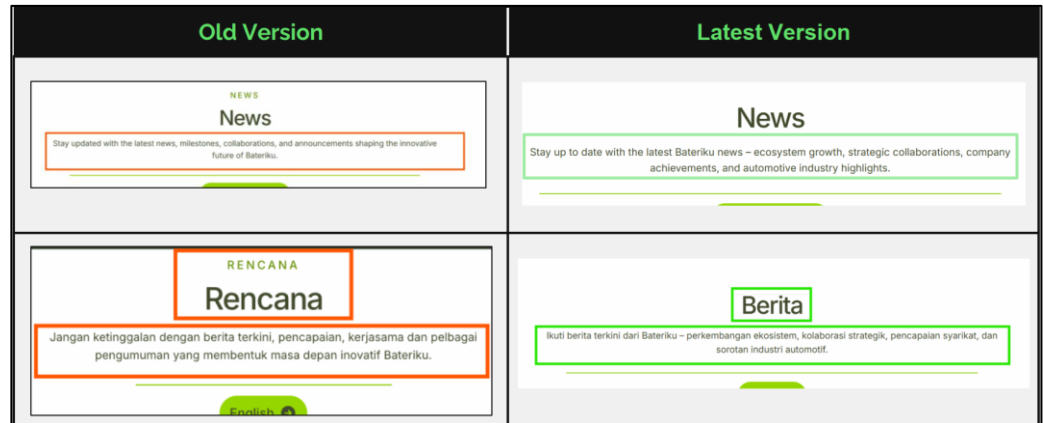


Figure 15: News page before and after update (business.bateriku.com)

2.3 Task 3: Structured Data Updates for bateriku.edu.my

2.3.1 Introduction

This is the Akademi Bateriku educational platform where they offer training courses and certification programs for automotive professionals. This task focused on improving search engine optimization (SEO) and information accuracy on bateriku.edu.my through structured data updates.

2.3.2 Implementation

The Battery Electric Vehicle (BEV) course page on bateriku.edu.my required structured data updates. The existing structured data was reviewed and updated to use the Course schema type with all recommended properties properly.

```

153 154 - "title": "Intensive Training",
155 155 + "title": "Intensive Training",
156 156 - "icon": "logicontraining",
157 157 + "description": "Hands-on training with modern equipment and real-world scenarios"
158 158 - },
159 159 + {
160 160 - "id": 4,
161 161 + "title": "Complete supervision",
162 162 + "title": "Complete supervision",
163 163 - "icon": "logiconsupervisor",
164 164 + "description": "Dedicated supervision and support from experienced professionals"
165 165 - },
166 166 + {
167 167 - "title": "Duration: 1-3 days",
168 168 + "title": "RPO",
169 169 - "icon": "logiconoutline",
170 170 + "description": "Choose a schedule that fits your lifestyle and work commitments"
171 171 - },
172 172 + "outcomes": [
173 173 - "Be able to describe the electric vehicle system functions and layout",
174 174 - "Perform basic precaution steps to work on electric vehicle (EV)",
175 175 + "Be able to describe the electric vehicle system functions and layout",
176 176 + "Perform basic precaution steps to work on electric vehicle (EV)",
177 177 - "Perform basic inspection for electric vehicle system and components according to industrial standard",
178 178 - "Perform preventive maintenance & diagnose malfunctions of high voltage (HV) battery systems and electrical and electronic systems related to electric vehicles",
179 179 + "Testing and ensuring the safety of electric vehicles can be used"
180 180 - ],
181 181 + "topics": [
182 182 - { "name": "Basic knowledge of electric vehicle (EV) system operation & function", "icon": "logiconlightning" },
183 183 - { "name": "Diagnosis & troubleshooting electric vehicle system", "icon": "logiconlightning" },
184 184 - { "name": "Power inverter system operation & troubleshooting", "icon": "logiconlightning" },
185 185 + { "name": "Diagnosis & troubleshooting electric vehicle system", "icon": "logiconlightning" },
186 186 + { "name": "Power drivetrain motor diagnosis & repair", "icon": "logiconlightning" }
187 187 + ]
188 188 - }
189 189 + }

```

Figure 16(a): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my). Red: before; Green: after.

```

188 188 - },
189 189 + "sections": [
190 190 - {
191 191 - "id": "electrical-vehicle-technology",
192 192 - "title": "",
193 193 - "theory": [
194 194 - "Basic principles of electric vehicle",
195 195 - "Type of electric vehicle mode",
196 196 - "Basic principles of electric vehicle",
197 197 - "Electric vehicle awareness",
198 198 - "Detailed explanation of how to measure voltage, resistance, and amperage using a multi-meter know how to work safely area electric vehicle",
199 199 - "Electric vehicle system component & functioning",
200 200 - "High voltage safety",
201 201 - "Isolate the electric vehicle process",
202 202 - "Identify electric vehicle isolate process",
203 203 - "Inspect HV inverter system using the multi-meter",
204 204 - "Type of electric vehicle mode",
205 205 - "Understand the danger of electric vehicle",
206 206 - "Know how to work safely area electric vehicle",
207 207 - "Comparison's hybrid electric vehicle & electric vehicle",
208 208 - "Electric vehicle system component & functioning",
209 209 - "Safety measures that need to be followed on electric vehicle",
210 210 - "High voltage safety",
211 211 - "Identify of safety equipment",
212 212 - "Isolate the electric vehicle process",
213 213 - "Understand of electric vehicle configuration",
214 214 - "Identify electric vehicle isolate process",
215 215 - "Checking high voltage batteries and battery service",
216 216 - "Inspect HV inverter system using the multi-meter",
217 217 - "Checking high voltage batteries and battery service"
218 218 - ]
219 219 + }
220 220 - }
221 221 + }

```

Figure 16(b): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my)

```

204 204 + "Checking high voltage batteries and battery service"
205 205 - },
206 206 - "practical": [
207 207 - "Using safety equipment",
208 208 - "How to check safety equipment",
209 209 - "Disable power source according to the procedure",
210 210 - "Disconnect HV service plug and auxiliary battery",
211 211 - "Checking HV battery voltage",
212 212 - "Checking auxiliary battery voltage",
213 213 - "Safety measures before dismantling HV battery",
214 214 - "Disconnect (HV) service plug from battery",
215 215 - "Remove & installation HV cable process",
216 216 - "Remove & installation HV cable process",
217 217 - "Inspect & check condition HV cable",
218 218 - "How to check safety equipment",
219 219 - "Disconnect HV service plug and auxiliary battery",
220 220 - "Checking auxiliary battery voltage",
221 221 - "Disconnect (HV) service plug from battery",
222 222 - "Remove & installation HV cable process",
223 223 - "Jump-start procedure process"
224 224 - ]
225 225 - }
226 226 - }

```

Figure 16(c): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my)

200	+	"Checking high voltage batteries and battery service"
205	205	},
207	207	"practical": {
208	+	"Using safety equipment",
209	+	"How to check safety equipment",
209	209	"Disable power source according to the procedure",
210	-	"Locating HV service plug and auxiliary battery",
211	209	"Checking HV battery voltage",
212	-	"Checking auxiliary battery voltage",
213	210	"Safety measures before dismantling HV battery",
214	-	"Disconnect (HV) service plug from battery",
215	211	"Remove & installation inverter process",
216	-	"Remove & installation HV cable process",
217	212	"Inspect & check condition HV cable",
218	+	"How to check safety equipment",
214	-	"Locating HV service plug and auxiliary battery",
215	+	"Checking auxiliary battery voltage",
216	-	"Disconnect (HV) service plug from battery",
217	+	"Remove & installation HV cable process",
218	215	"Jump-start procedure process"
219	219	}
220	220	}

Figure 16(d): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my)

```

284 +         "Checking High voltage batteries and battery service"
285 +     },
286 +     "practical": [
287 +         "Using safety equipment",
288 +         "How to check safety equipment",
289 +         "Disable power source according to the procedure",
290 +         "Locating HV service plug and auxiliary battery",
291 +         "Checking HV battery voltage",
292 +         "Checking auxiliary battery voltage",
293 +         "Safety measures before dismantling HV battery",
294 +         "Disconnect (OV) service plug from battery",
295 +         "Remove & installation inverter process",
296 +         "Remove & installation HV cable process",
297 +         "Inspect & check condition HV cable",
298 +         "How to check safety equipment",
299 +         "Locating HV service plug and auxiliary battery",
300 +         "Checking auxiliary battery voltage",
301 +         "Disconnect (OV) service plug from battery",
302 +         "Remove & installation HV cable process",
303 +         "Jump-start procedure process"
304 +     ],
305 + }
306 +
307 +
308 +
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```

Figure 16(e): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my)

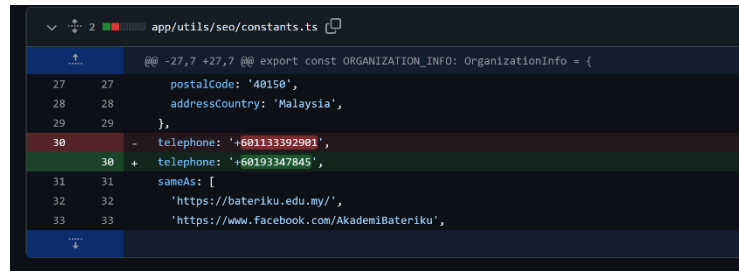
```

8 @@ -7,7 +7,7 @@ export const SID_DEFAULTS = {
9   batteryImage: 'bakademiatsterku',
10  separator: '-',
11  defaultPage: '/seo.ng.png',
12  fallbackDescription: "Akademi Bateriku Sdn. Bhd. provides hands on automotive training in ICE & EV, with 3PK SLON certified programs designed to equip you with industry-ready skills",
13  fallbackDescription: "Akademi Bateriku Sdn. Bhd. provides hands on automotive training in ICE & EV, with 3PK SLON certified programs designed to equip you with industry-ready skills",
14  defaultKeywords: "car battery, bateriku, auto services, emergency battery replacement",
15  defaultAuthor: "bateriku.edu.my",
16  defaultType: "website" as const,
17 }
18
19 @ -19,19 +19,22 @@ export const ORGANIZATION_INFO: OrganizationInfo = {
20   url: "https://bateriku.edu.my/",
21   logo: "/favicon-256x256.png",
22   image: "/seo.ng.png",
23   description: "Akademi Bateriku Sdn. Bhd. provides hands on automotive training in ICE & EV, with 3PK SLON certified programs designed to equip you with industry-ready skills",
24   description: "Akademi Bateriku Sdn. Bhd. provides hands on automotive training in ICE & EV, with 3PK SLON certified programs designed to equip you with industry-ready skills",
25   address: {
26     streetAddress: "No 15, Jalan Sawjana Indah 1B, Taman Perindustrian Sawjana Indah TPOJ Jaya, Sekyen 02",
27     addressLocality: "Shah Alam",
28     addressRegion: "Selangor",
29     postalCode: "40158",
30     addressCountry: "MY",
31     addressCountry: "Malaysia",
32   },
33   telephone: "+601133392961",
34   socialMedia: [
35     "https://bateriku.edu.my/",
36     "https://www.facebook.com/AkademiBateriku/",
37     "https://www.instagram.com/AkademiBateriku/?hl=en",
38     "https://x.com/baterikudotcom",
39     "https://www.instagram.com/AkademiBateriku/?hl=en",
40     "https://www.youtube.com/channel/UCGh3Zm6dRwCzmgqtwr",
41     "https://www.linkedin.com/company/baterikudotcom",
42   ],
43 };

```

Figure 16(f): Code for updating structured data (@type: Course) for BEV course (bateriku.edu.my)

Moreover, the Organization's structured data on bateriku.edu.my was updated to replace the old phone number with the current contact information.



```

app/utis/seo/constants.ts
@@ -27,7 +27,7 @@ export const ORGANIZATION_INFO: OrganizationInfo = {
 27 27     postalCode: '40150',
 28 28     addressCountry: 'Malaysia',
 29 29 },
30 -   telephone: '+601133392901',
31 +   telephone: '+60193347845',
32   sameAs: [
33     'https://bateriku.edu.my/',
     'https://www.facebook.com/AkademiBateriku',
  
```

Figure 17: Code for updating Organization structured data on Akademi Bateriku (bateriku.edu.my). Red: before; Green: after.

2.3.3 Results and Discussion

After the update, the website reflected accurate course and organization information, improving SEO performance and ensuring users receive correct contact details.

Course		0 ERRORS 0 WARNINGS
ID: https://bateriku.edu.my/courses/battery-electric-vehicle-bev#Course		
@type	Course	
@id	https://bateriku.edu.my/courses/battery-electric-vehicle-bev#Course	
name	Battery Electric Vehicle (BEV)	
about		
@type	Thing	
name	Battery Electric Vehicle (BEV)	
hasPart		
@type	CreativeWork	
name	Recognized by JPK	
hasPart		
@type	CreativeWork	
name	Complete Guidance	
hasPart		
@type	CreativeWork	
name	Intensive Training	
hasPart		
@type	CreativeWork	
name	Complete Supervision	
hasPart		
@type	CreativeWork	
name	P3D	
hasPart		
@type	LearningResource	
name	What You Will Learn	
hasPart		
@type	LearningResource	
name	Module 1: Be able to describe the Electric Vehicle System functions	

Figure 18: Structure data schema of BEV course page after update (bateriku.edu.my)

Old Version	Latest Version																																						
Organization 0 ERRORS 0 WARNINGS ID: https://www.bateriku.edu.my/en#org <table> <tr><td>@type</td><td>Organization</td></tr> <tr><td>@id</td><td>https://www.bateriku.edu.my/en#org</td></tr> <tr><td>name</td><td>Akademi Bateriku Sdn Bhd</td></tr> <tr><td>uri</td><td>https://bateriku.edu.my/</td></tr> <tr><td>logo</td><td>https://www.bateriku.edu.my/favicon-256x256.png</td></tr> <tr><td>image</td><td>https://www.bateriku.edu.my/seo_bg.png</td></tr> <tr><td>description</td><td>Akademi Bateriku Sdn. Bhd. provides hands on automotive training in ICE & EV, with JPK SLDN certified programs designed to equip you with industry-ready skills.</td></tr> <tr><td>telephone</td><td>+60133392901</td></tr> <tr><td>sameAs</td><td>https://bateriku.edu.my/</td></tr> <tr><td>sameAs</td><td>https://www.facebook.com/AkademiBateriku</td></tr> </table>	@type	Organization	@id	https://www.bateriku.edu.my/en#org	name	Akademi Bateriku Sdn Bhd	uri	https://bateriku.edu.my/	logo	https://www.bateriku.edu.my/favicon-256x256.png	image	https://www.bateriku.edu.my/seo_bg.png	description	Akademi Bateriku Sdn. Bhd. provides hands on automotive training in ICE & EV, with JPK SLDN certified programs designed to equip you with industry-ready skills.	telephone	+60133392901	sameAs	https://bateriku.edu.my/	sameAs	https://www.facebook.com/AkademiBateriku	Organization 0 ERRORS 0 WARNINGS ID: https://www.bateriku.edu.my/en#org <table> <tr><td>@type</td><td>Organization</td></tr> <tr><td>@id</td><td>https://www.bateriku.edu.my/en#org</td></tr> <tr><td>name</td><td>Akademi Bateriku Sdn Bhd</td></tr> <tr><td>uri</td><td>https://bateriku.edu.my/</td></tr> <tr><td>logo</td><td>https://www.bateriku.edu.my/favicon-256x256.png</td></tr> <tr><td>image</td><td>https://www.bateriku.edu.my/seo_bg.png</td></tr> <tr><td>description</td><td>Akademi Bateriku Sdn. Bhd. provides hands on automotive training in ICE & EV, with JPK SLDN certified programs designed to equip you with industry-ready skills.</td></tr> <tr><td>telephone</td><td>+60193347845</td></tr> <tr><td>sameAs</td><td>https://bateriku.edu.my/</td></tr> </table>	@type	Organization	@id	https://www.bateriku.edu.my/en#org	name	Akademi Bateriku Sdn Bhd	uri	https://bateriku.edu.my/	logo	https://www.bateriku.edu.my/favicon-256x256.png	image	https://www.bateriku.edu.my/seo_bg.png	description	Akademi Bateriku Sdn. Bhd. provides hands on automotive training in ICE & EV, with JPK SLDN certified programs designed to equip you with industry-ready skills.	telephone	+60193347845	sameAs	https://bateriku.edu.my/
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image	https://www.bateriku.edu.my/seo_bg.png																																						
description	Akademi Bateriku Sdn. Bhd. provides hands on automotive training in ICE & EV, with JPK SLDN certified programs designed to equip you with industry-ready skills.																																						
telephone	+60193347845																																						
sameAs	https://bateriku.edu.my/																																						

Figure 19: Structure data schema of Organization after update (bateriku.edu.my)

2.4 Task 4: Image Fix for corporate.bateriku.com

2.4.1 Introduction

This website contains corporate information about the company, including company news, milestones, and information. This task involved fixing a visual display issue on corporate.bateriku.com, specifically related to image consistency on the News page.

2.4.2 Implementation

The issue was caused by inconsistent image sizing due to styling constraints. CSS adjustments were made to ensure images were displayed uniformly across the News page.

```

1 1 <template>
2 2   <component :is="ImageComponent" placeholder="/compress/no-image.png" class="img-4 rounded-3xl" :src="refinedSrc" :alt="props.alt" :width="props.width" :height="props.height" />
3 3   <component :is="ImageComponent" placeholder="/compress/no-image.png" class="img-4 rounded-3xl" :src="refinedSrc" :alt="props.alt" :width="props.width" :height="props.height" />
4 4   </template>
5 5   <script setup lang="ts">

```

Figure 20: Code for fixing inconsistent image display on Corporate Bateriku News page (corporate.bateriku.com). Red: before; Green: after.

2.4.3 Results and Discussion

The images now appear consistent in size and alignment, resulting in a more professional and visually appealing layout.

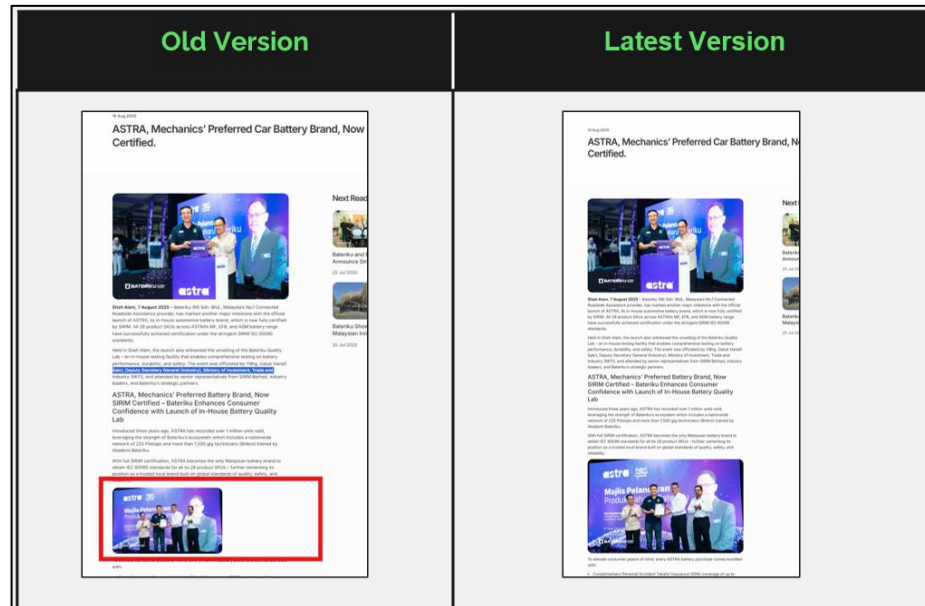


Figure 21: Image display before and after fix (corporate.bateriku.com)

2.5 Task 5: Frontend Development & SEO Implementation for astraman.my

2.5.1 Introduction

This website, which is one of Bateriku's related brands, focusing on automotive products including BForty Series and Astra Series batteries. This task involved frontend development and SEO improvements for astraman.my, including successful and unsuccessful pages, structured data updates, and creating a sitemap entry for the News page. All tasks were implemented in the staging environment and have not yet been deployed to production.

2.5.2 Implementation

The following tasks were implemented:

- Developed successful (thank you) and unsuccessful pages for Contact Us and Sponsorship forms based on Figma design.

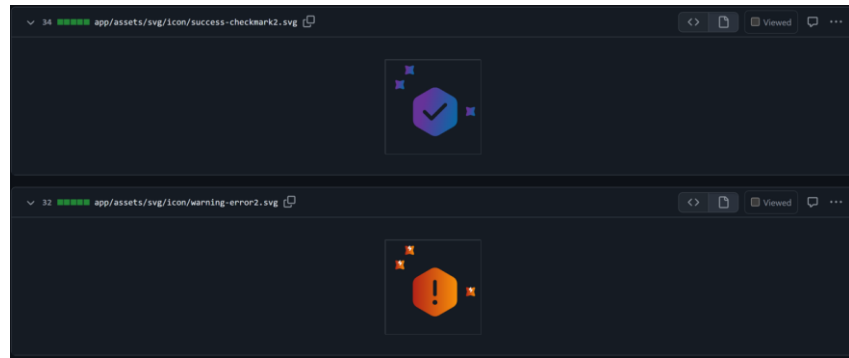


Figure 22: Icon images used for successful and unsuccessful pages (astraman.my)

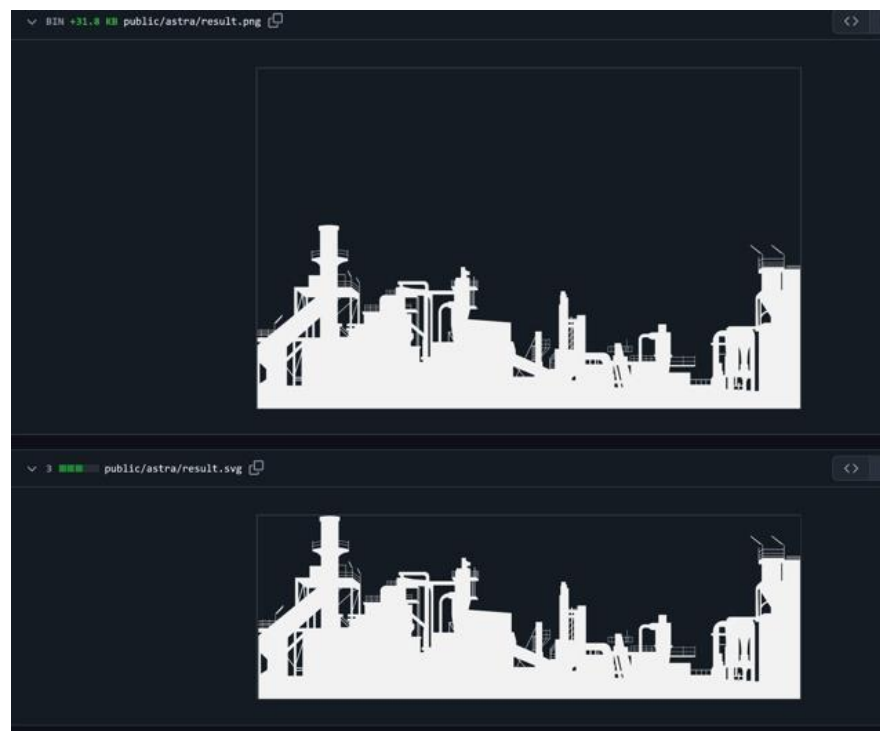


Figure 23: Background image used for successful and unsuccessful pages (astraman.my)

```

95 + </style>
68 app/pages/result/success.vue
... @@ -0,0 +1,68 @@
1 + <template>
2 + <main class="pt-[6.8rem]">
3 + <section id="hero" data-section="white" class="relative py-10 sm:py-[6.25rem] px-5 lg:px-16 xl:px-20">
4 + 
6 + <div class="mx-auto max-w-7xl relative h-full">
7 + <div class="intersect-once intersect:animate-fade-d">
8 + <!-- Icon Section -->
9 + <div class="flex flex-col justify-center items-center px-6 pt-20 pb-6">
10 + <!-- Success Icon with gradient background -->
11 + <img alt="Success icon" class="text-9xl :fontControlled=true :filled=true" />
12 + </div>
13 + <!-- Title -->
14 + <div class="px-3 sm:px-8 pb:4">
15 + <h2
16 + class="text-6xl font-semibold tracking-tight antialiased text-center bg-[linear-
17 + gradient(103.45deg, #663399_25.98%, #66CA7_96.21%)] bg-clip-text leading-normal text-transparent pb-2">
18 + Received! <br />Thank you for Reaching Out.
19 + </h2>
20 + </div>
21 + <div class="intersect-once intersect:animate-bounce-u">
22 + <!-- Description -->
23 + <div class="mt-8 text-center mx-auto max-w-[900px] px-3 sm:px-8">
24 + <p class="text-lg font-normal antialiased text-[#414042] leading-7">
25 + We've received your message and our team will get back to you as soon as possible. In the meantime, feel
26 + free to explore more of what we offer.
27 + </p>
28 + </div>
29 + <!-- Button Section -->
30 + <div class="px-6 py-12 flex text-center place-self-center">
31 + <button @click="navigateTo('/')" type="button"
32 + class="w-64 md:w-[280px] px-4 py-3.5 bg-[#663399] rounded-[100px] text-base font-medium text-[#FEFEFE] tracking-[0.01em] hover:bg-
33 + #5a2d8a transition-colors duration-200">
34 + Back to Home
35 + </button>
36 + </div>
37 + </div>
38 + </section>
39 + </main>
40 + </template>
41 + <script lang="ts" setup>
42 + import { LazySvgIconSuccessCheckmark2 } from '#components';
43 +
44 + defineOptions({
45 + name: "ResultSuccess",
46 + });
47 +
48 + definePageMeta({
49 + layout: "astra"
50 + });
51 +
52 + defineRoute({
53 + paths: {
54 + en: "/result/success",
55 + },
56 + });
57 +
58 + useSectionIntersect();
59 + </script>
60 + <style></style>

```

Figure 24: Source code implementation for the successful page (astraman.my)

```

95 app/pages/result/failed.vue
... @@ -0,0 +1,95 @@
1 + <template>
2 +   <main class="pt-[6.8rem]">
3 +     <section id="hero" data-section="white" class="relative py-10 sm:py-[6.25rem] px-5 lg:px-16 xl:px-20">
4 +       <NextImg form="webp" loading="lazy"
5 +         class="absolute h-auto mx-h-[900px] w-full object-cover bottom-0 bg-bottom object-center mx-auto left-0 -z-10"
6 +         src="/astra/result.svg" alt="result bg"/>
7 +       <div class="mx-auto max-w-7xl relative h-full">
8 +
9 +         <div class="intersect-once intersect:animate-fade-d">
10 +           <!-- Icon Section -->
11 +           <div class="flex flex-col justify-center items-center px-6 pt-20 pb-6">
12 +             <!-- Success Icon with gradient background -->
13 +             <LazySvgIconWarningError2 class="text-9xl" :fontControlled="true" :filled="true"/>
14 +           </div>
15 +
16 +           <!-- Title -->
17 +           <div class="px-3 sm:px-8 pb-4">
18 +             <h2 class="text-6xl font-semibold tracking-tight antialiased text-center bg-[linear-gradient(90deg,_#B42318_0%,_#F79009_100%)]
19 +               bg-clip-text leading-normal text-transparent pb-2">
20 +               Uh-oh! <br />Something went wrong.
21 +             </h2>
22 +           </div>
23 +
24 +           <div class="intersect-once intersect:animate-bounce-u">
25 +             <!-- Description -->
26 +             <div class="mt-8 text-center mx-auto max-w-[900px] px-4 sm:px-8">
27 +               <p class="text-lg font-normal antialiased text-[#414042] leading-7">
28 +                 We couldn't send your message. <br />Please check your details and try again, or contact us directly if the issue persists
29 +               </p>
30 +             </div>
31 +
32 +             <!-- Button Section -->
33 +             <div class="px-4 py-12 flex flex-col md:flex-row text-center place-self-center gap-8 opacity-100">
34 +               <button
35 +                 @click="navigateTo('/')"
36 +                 type="button"
37 +                 class="w-64 md:w-[280px] px-4 py-3.5 border-2 border-[#663399] bg-white rounded-[100px] text-base sm:text-lg font-medium
38 +                 text-[#663399] tracking-[0.01em] hover:bg-[#F8F7F7] transition-colors duration-200"
39 +               >
40 +                 Back to Home
41 +               </button>
42 +
43 +               <button
44 +                 @click="handleTryAgain()"
45 +                 type="button"
46 +                 class="w-64 md:w-[280px] px-4 py-3.5 bg-[#663399] rounded-[100px] text-base sm:text-lg font-medium text-[#FEFEFE] tracking-
47 +                 [0.01em] hover:bg-[#5A2D8A] transition-colors duration-200"
48 +               >
49 +                 Try Again
50 +               </button>
51 +             </div>
52 +           </div>
53 +         </section>
54 +       </main>
55 +     </template>
56 +
57 +     <script lang="ts" setup>
58 +       <import {LazySvgIconWarningError2} from '#components';
59 +
60 +       <defineOptions({
61 +         name: "ResultFailed",
62 +       })>
63 +
64 +       <definePageMeta({
65 +         layout: "astra"
66 +       })>
67 +
68 +       <defineI18nRoute({
69 +         paths: {
70 +           en: "/result/failed",
71 +         },
72 +       })>
73 +
74 +       <useSectionIntersect();
75 +
76 +       <const route = useRoute();
77 +       <const router = useRouter();
78 +
79 +       <const handleTryAgain = () => {
80 +         <const from = route.query.from;
81 +         <if (from === 'contact') {
82 +           <navigateTo('/contact');
83 +         } <else if (from === 'sponsorship') {
84 +           <navigateTo('/sponsorship');
85 +         } <else {
86 +           <router.go(-1);
87 +         }
88 +       };
89 +
90 +     </script>
91 +
92 +     <style>
93 +
94 +     </style>
95 +

```

Figure 25: Source code implementation for the unsuccessful page (astraman.my)

- Updated structured data thumbnails for Sponsorship and Contact Us pages.
- Created a post entry in the sitemap for the News page.

```

+ news: {
+   title: p.title,
+   publication_date: p.publishedAt ? (isNaN(Date.parse(p.publishedAt)) ? null : new Date(p.publishedAt).toISOString()) : null,
+   publication: {
+     name: "Business Updates",
+     language: "en",
+   },
+ },
+ _sitemap: "posts",
+ )))

```

Figure 26: Source code for creating the post entry in the sitemap (astraman.my)

- Implemented structured data for the Product Catalogue.

```

53  app/pages/catalogue/index.vue
@@ -98,6 +98,25 @@ const { data: catalogues } = await useAsyncData('catalogues', () =>
98  98  onMounted(async () => {
99  99  // fetch vehicle model and manufacturers on page load
100 100  await store.fetchVehicleBrands()

101  +
102  + useHead(() => {
103  +   return generateSEO({
104  +     title: "Products - Astra Battery Catalogue",
105  +     description:
106  +       "Browse our complete range of high-performance Astra batteries.",
107  +     image: `${requestURL.origin}/img/thumbnail/product-catalogue.png`,
108  +     organizationImage: `${requestURL.origin}/seo/homepage.png`,
109  +     url: `${requestURL.origin}${requestURL.pathname}`,
110  +     catalogues: (catalogues.value || []).map(catalogue => ({
111  +       name: catalogue.name,
112  +       aggregateRating: {
113  +         "@type": "AggregateRating",
114  +         ...getDeterministicRatingFromId(catalogue.id)
115  +       }
116  +     )))
117  + })

```

Figure 27: Source code for implementing structured data for the Product Catalogue (astraman.my)

2.5.3 Results and Discussion

The implemented pages now provide clear feedback to users after form submission. The structured data and sitemap updates improved SEO visibility and ensured accurate content indexing by search engines.

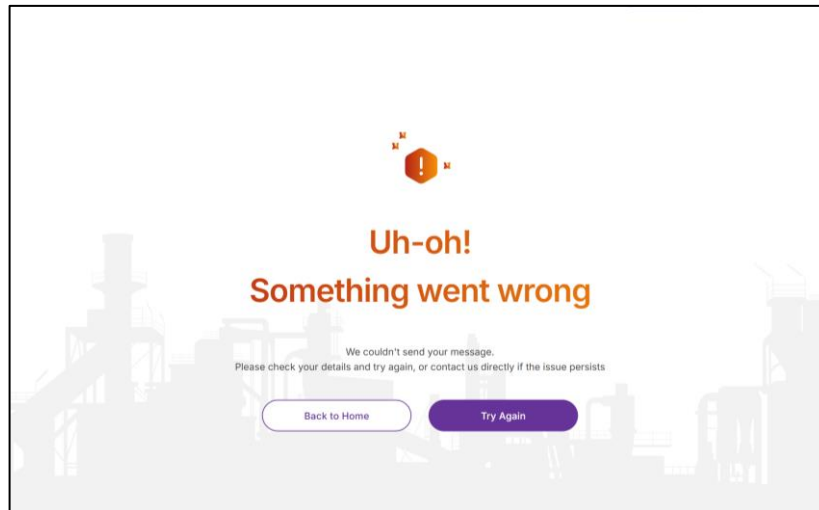


Figure 28: Unsuccessful page (astraman.my)

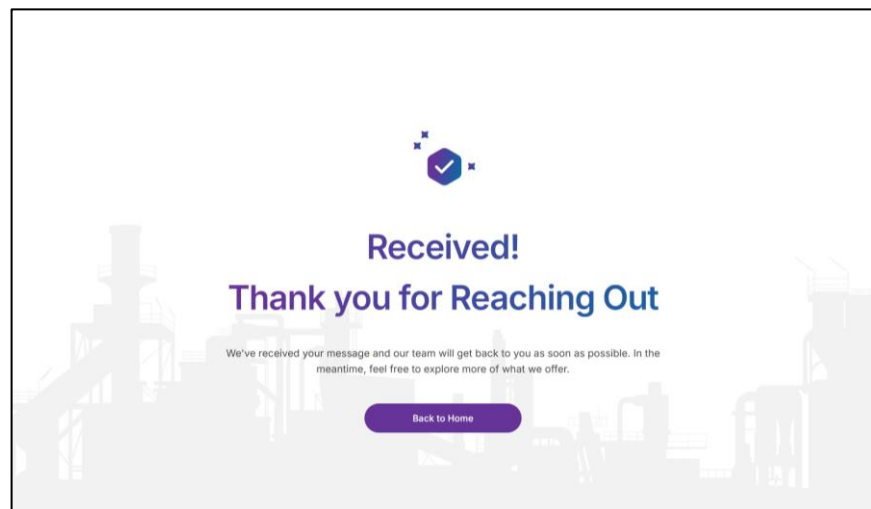


Figure 29: Successful(thank you) page (astraman.my)

	Old Version	Latest Version
Sponsorship		
Contact Us		

Figure 30: Structured data thumbnails before and after update for the Sponsorship and Contact Us pages (astraman.my)

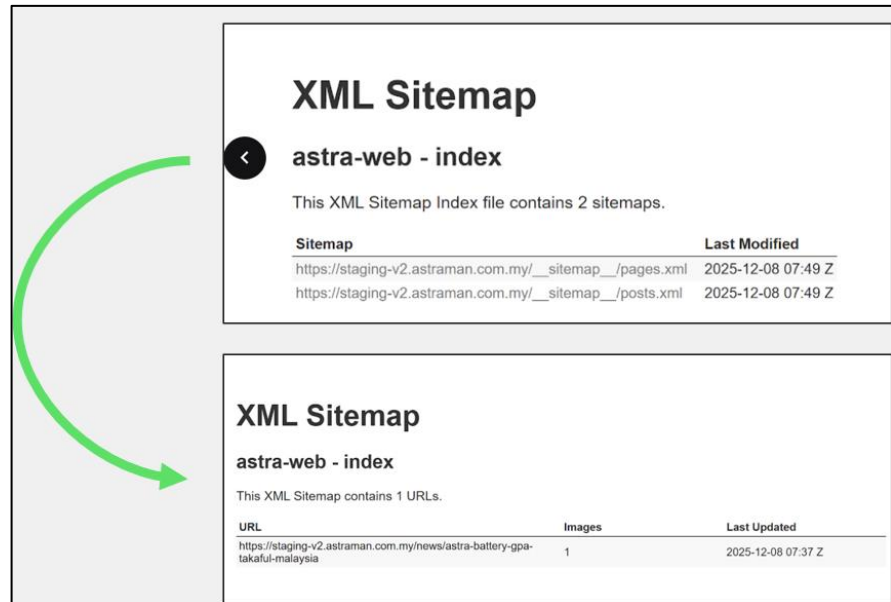


Figure 31: Post entry in the sitemap for the News page (astraman.my)

WebPage	0 ERRORS	0 WARNINGS	1 ITEM
Organization	0 ERRORS	0 WARNINGS	1 ITEM
Product	0 ERRORS	0 WARNINGS	11 ITEMS
Product	0 ERRORS	0 WARNINGS	^
@type	Product		
name	DIN60 (LN2) EFB ASTRA (SB)		
aggregateRating			
@type	AggregateRating		
ratingValue	4.5		
reviewCount	50		
Product	0 ERRORS	0 WARNINGS	^
@type	Product		
name	NS40ZL (40B19L) MF BFORTY		
aggregateRating			
@type	AggregateRating		
ratingValue	4.5		
reviewCount	50		
Product	0 ERRORS	0 WARNINGS	^
@type	Product		
name	Astra NS70L		
aggregateRating			

Figure 32: Structured data schema for the Product Catalogue (astraman.my)

CHAPTER 3

CONCLUSION AND RECOMMENDATIONS

3.1 Benefit of the assignments

During my industrial training at Bateriku.com, multiple tasks were assigned across several company websites, including bateriku.com, business.bateriku.com, bateriku.edu.my, corporate.bateriku.com, and astraman.my. These assignments provided valuable opportunities to enhance my technical skills. I gained practical experience in frontend development using the Nuxt framework and Visual Studio Code, where I updated website content, fixed layout issues, implemented structured data for SEO, and improved the overall user experience.

The tasks also helped me develop my problem-solving abilities. Many assignments required me to troubleshoot code errors, resolve layout overlaps, and test website functionality across multiple platforms. I learned to analyze problems carefully, apply solutions, and verify the results effectively.

Working closely with the production team, my mentor Encik Aiman, and other frontend team members gave me a better understanding of professional workflow. I learned how to clarify task requirements, follow instructions accurately, and complete assignments within deadlines. Moreover, working with Figma designs for Astraman pages taught me how to translate design specifications into functional web pages, which improved my practical understanding of web development in a professional environment.

3.2 Summary of Industrial Training

Overall, my industrial training at Bateriku.com has been a highly rewarding experience that allowed me to bridge the gap between academic knowledge and real-world application. Through this training, I enhanced my technical skills in web development, coding, and website maintenance, while also improving my

communication skills by interacting with team members, mentor, and the production team.

The experience strengthened my problem-solving and critical thinking skills, as I learned to handle unexpected issues and complete tasks efficiently. I also gained insight into professional workplace standards and team collaboration, which are essential for a career in the IT industry. Furthermore, managing multiple websites and tasks helped me improve my time management and build confidence in handling complex assignments.

In conclusion, this industrial training has provided me with a solid foundation for my future career in web development and multimedia computing, while giving me a better understanding of professional work ethics, teamwork, and technical problem-solving in a real industry setting.

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